

YONGGANG JIANG

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EDUCATION

Max Planck Institute for Informatics

Apr. 2023 – Present

PhD in Computer Science

Advisor: [Danupon Nanongkai](#)

Max Planck Institute for Informatics

Oct. 2021 – Mar. 2023

Preparatory Phase for Doctoral Study

Nanjing University

Sep. 2017 – Jul. 2021

B.S. in Computer Science and Technology

GPA: 94.4/100, Ranking: 1/31

University of California, Berkeley

Jan. 2020 – May 2020

Berkeley International Study Program

GPA: 4.0/4.0

HONORS & AWARDS

Google PhD Fellowship

2025

Distinguished Paper Award, SPAA 2025

2025

Outstanding Graduate, Nanjing University

2021

National Elite Program Scholarship, Outstanding Prize (top 1)

2018, 2019, 2020

China Collegiate Programming Contest (CCPC), Gold Medal

2018

National Olympiad in Informatics in Provinces (NOIP), First Prize

2015

RESEARCH INTERESTS

I work broadly at the intersection of algorithms and mathematics in computer science.

Current focus

- Design and analysis of graph algorithms.
- Theory of parallel and distributed computing.

Active explorations

- Automated formalization and theorem proving (Lean, LLM-assisted)
- Practice-driven problems involving graph algorithms and high-performance implementations.

PUBLICATIONS

* Authors are listed alphabetically with **equal contribution** (see [convention in mathematics](#)).

Manuscripts under review

- [1] **Deterministic Negative-Weight Shortest Paths in Nearly Linear Time via Path Covers**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
- [2] **DAG Projections: Reducing Distance and Flow Problems to DAGs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.

- [3] **A Subquadratic Two-Party Communication Protocol for Minimum Cost Flow**
Hossein Gholizadeh, Yonggang Jiang.
- [4] **Parallel Small Vertex Connectivity in Near Linear Work and Polylogarithmic Depth**
Yonggang Jiang, Changki Yun.
- [5] **Reviving Thorup's Shortcut Conjecture**
Aaron Bernstein, Henry Fleischmann, George Z. Li, Bernhard Haeupler, Gary Hoppenworth, Yonggang Jiang, Maximilian Probst Gutenberg, Seth Pettie, Thatchaphol Saranurak, Leon Schiller.

Published

- [1] **Perfect Simulation of Las Vegas Algorithms via Local Computation**
Xinyu Fu, Yonggang Jiang, Yitong Yin.
Innovations in Theoretical Computer Science (ITCS) 2026.
- [2] **Reducing Shortcut and Hopset Constructions to Shallow Graphs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.
- [3] **Directed and Undirected Vertex Connectivity Problems are Equivalent for Dense Graphs**
Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.
- [4] **Minimum s-t Cuts with Fewer Cut Queries**
Danupon Nanongkai, Yonggang Jiang, Pachara Sawettamalya.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.
- [5] **Shortcuts and Transitive-Closure Spanners Approximation**
Parinya Chalermsook, Yonggang Jiang, Sagnik Mukhopadhyay, Danupon Nanongkai.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.
- [6] **New Oracles and Labeling Schemes for Vertex Cut Queries**
Yonggang Jiang, Merav Parter, Asaf Petruschka.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.
- [7] **Parallel $(1 + \epsilon)$ -Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work**
Bernhard Haeupler, Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Shengzhe Wang.
IEEE Symposium on Foundations of Computer Science (FOCS) 2025.
- [8] **Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances**
Jan van den Brand, Hossein Gholizadeh, Yonggang Jiang, Tijn de Vos.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2025.
Distinguished Paper Award
- [9] **Deterministic Vertex Connectivity via Common-Neighborhood Clustering and Pseudorandomness**
Chaitanya Nalam, Yonggang Jiang, Thatchaphol Saranurak, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.
- [10] **Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations**
Joakim Blikstad, Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.

- [11] **Parallel and Distributed Exact Single-Source Shortest Paths with Negative Edge Weights**
Vikrant Ashvinkumar, Aaron Bernstein, Nairen Cao, Christoph Grunau, Bernhard Haeupler,
Yonggang Jiang, Danupon Nanongkai, Hsin Hao Su.
European Symposium on Algorithms (ESA) 2024.
- [12] **Finding a Small Vertex Cut on Distributed Networks**
Yonggang Jiang, Sagnik Mukhopadhyay.
ACM Symposium on Theory of Computing (STOC) 2023.
- [13] **Robust and Optimal Contention Resolution without Collision Detection**
Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2022.
- [14] **Tight Trade-off in Contention Resolution without Collision Detection**
Haimin Chen, Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Principles of Distributed Computing (PODC) 2021.

RESEARCH VISITS & INVITED TALKS

National University of Singapore , Singapore, Hosted by Yi-Jun Chang. Talk: Parallel $(1 + \epsilon)$ -Approximate Mincost Flow in Almost Optimal Depth and Work	<i>Sep. 2025</i>
New York University , New York, NY, USA Hosted by Aaron Bernstein. Talk: Parallel $(1 + \epsilon)$ -Approximate Mincost Flow in Almost Optimal Depth and Work	<i>Sep. 2025</i>
Carnegie Mellon University , Pittsburgh, PA, USA Hosted by Jason Li. Talk: Parallel $(1 + \epsilon)$ -Approximate Mincost Flow in Almost Optimal Depth and Work	<i>Aug. 2025</i>
University of Michigan , Ann Arbor, MI, USA Hosted by Thatchaphol Saranurak.	<i>Jul. 2025</i>
ETH Zurich , Zurich, Switzerland Hosted by Sorrachai Yingchareonthawornchai.	<i>Feb. 2025</i>
Tsinghua University , Beijing, China Hosted by Ran Duan. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Dec. 2024</i>
Nanjing University , Nanjing, China Hosted by Yitong Yin. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Dec. 2024</i>
Shanghai University of Finance and Economics , Shanghai, China Hosted by Pinyan Lu. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Dec. 2024</i>
ETH Zurich , Zurich, Switzerland Hosted by Weiming Feng. Talk: New Tools and Faster Algorithms for Vertex Min-Cut	<i>Nov. 2024</i>
INSAIT , Sofia, Bulgaria Hosted by Bernhard Haeupler.	<i>Sep. 2024</i>
University of Warwick , Coventry, UK Hosted by Sayan Bhattacharya. Talk: Deterministic k-Vertex Connectivity in k Max-Flows	<i>Jun. 2024</i>

University of Sheffield , Sheffield, UK Hosted by Sagnik Mukhopadhyay.	<i>Jun. 2024</i>
Nanjing University , Nanjing, China Hosted by Penghui Yao.	<i>Nov. 2023</i>
Rutgers University , New Brunswick, NJ, USA Hosted by Aaron Bernstein.	<i>Jun. 2023</i>
Aalto University , Helsinki, Finland Hosted by Parinya Chalermsook.	<i>Apr. 2023</i>
Aalto University , Helsinki, Finland Hosted by Parinya Chalermsook.	<i>Dec. 2022</i>
University of Sheffield , Sheffield, UK Hosted by Sagnik Mukhopadhyay.	<i>May 2022</i>

CONFERENCE & WORKSHOP TALKS

Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances <i>SPAA 2025, Portland, OR, USA</i>	<i>Jul. 2025</i>
Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations <i>STOC 2025, Prague, Czech Republic</i>	<i>Jun. 2025</i>
Contributed Talk: Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations <i>HALG 2025, Zurich, Switzerland</i>	<i>Jun. 2025</i>
Contributed Talk: Deterministic k-Vertex Connectivity in k Max-Flows <i>HALG 2024, Warsaw, Poland</i>	<i>Jun. 2024</i>
Parallel, Distributed, and Quantum Exact Single-Source Shortest Paths with Negative Edge Weights <i>ESA 2024, London, UK</i>	<i>Jun. 2024</i>
Finding a Small Vertex Cut on Distributed Networks <i>STOC 2023, Orlando, FL, USA</i>	<i>Jun. 2023</i>
Robust and Optimal Contention Resolution without Collision Detection <i>SPAA 2022, online</i>	<i>Jul. 2022</i>
Tight Trade-off in Contention Resolution without Collision Detection <i>PODC 2021, online</i>	<i>Jul. 2021</i>

TEACHING EXPERIENCE

Randomized and Approximation Algorithms , MPI-INF Role: Teaching Assistant.	<i>Winter 2025</i>
Seminar: Foundations of Machine Learning , MPI-INF Role: Coordinator and Speaker.	<i>Summer 2023</i>
Data Structures and Algorithms , Nanjing University Role: Teaching Assistant.	<i>Fall 2020, Fall 2019</i>

STUDENTS & MENTORING

Ioannis Dorkofikis (M.Sc.)

Internship at MPI-INF, from NTUA, Greece.

Spring 2025

Changki Yun (B.Sc.)

Internship at MPI-INF, from SNU, Korea.

Outcome: [Paper on parallel vertex connectivity](#) (in submission)

Fall 2024

Hossein Gholizadeh (B.Sc.)

Bachelor's Thesis Internship at MPI-INF, from KIT, Germany.

Outcome: [Paper in SPAA 2025 \(Distinguished Paper Award\)](#)

Summer 2024

Alireza Danaei (B.Sc.)

Internship at MPI-INF, from Sharif Univ., Iran

Summer 2023

Vassilis Xanthopoulos (M.Sc.)

Master's Thesis Internship at MPI-INF, from NTUA, Greece

Spring 2023